Trigonometry to Find Angles

Exam code: 0607

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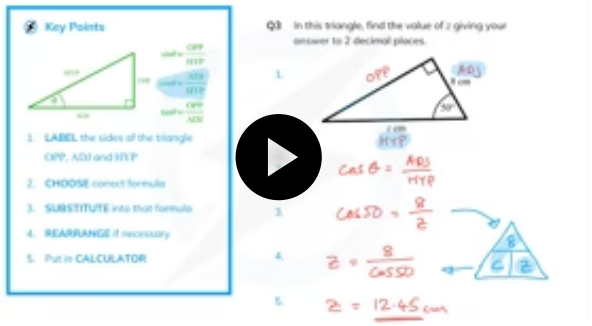

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SOHCAHTOA – Finding angles

Help me understand

How can I use SOHCAHTOA to find missing angles?

- STEP 1
  Label the sides of the triangle as H, O and A
  - **H** is the **longest side** opposite the right angle
  - **O** is **opposite** the given angle
  - **A** is **next to** the given angle
- STEP 2
  Identify which **trigonometric ratio** to use: sin, cos or tan
  - Write down the letters of the **lengths** you are **given**
  - Find the two letters in **SOHCAHTOA** to identify which ratio to use
    - If you have **O** and **A** then use **tan**
- STEP 3
  Substitute the values into the relevant trigonometric formula
  - The **angle** will be **unknown**
    - $\tan(\theta) = \frac{3}{4}$
- STEP 4
  Substitute the fraction into the **inverse trigonometric**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

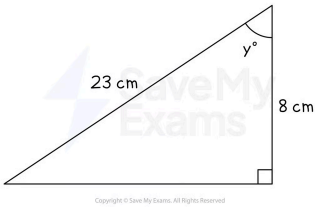
- $\tan(\theta) = \frac{3}{4}$ leads to $\theta = \tan^{-1}\left(\frac{3}{4}\right)$

- **STEP 5**
  Type the expression **into your calculator**
  - The **question** might ask you to **round your answer**
  - If not then round to **one decimal place**

Worked Example

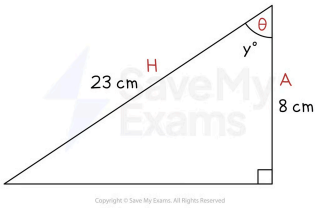
Find the value of the angle $y°$ in the following triangle.

Give your answer to 1 decimal place.



Answer:

First label the triangle



We know A and H - that's CAH or $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

$$\cos(y) = \frac{8}{23}$$

Use inverse cos to find y

$$y = \cos^{-1}\left(\frac{8}{23}\right)$$

Enter on your calculator

$$y = 69.6455...$$

Round to 1 decimal place

$$y = 69.6°$$

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